

JAVA · COMMAND-LINE APPLICATION

Bank of CLI

A command-line Java application built to show object-oriented design, clean separation of concerns, and persistence.

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STACK



Java 17 · JDBC · SLF4J



JUnit 5



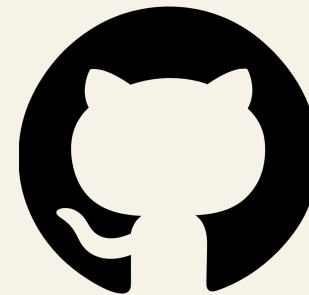
Mockito



Maven



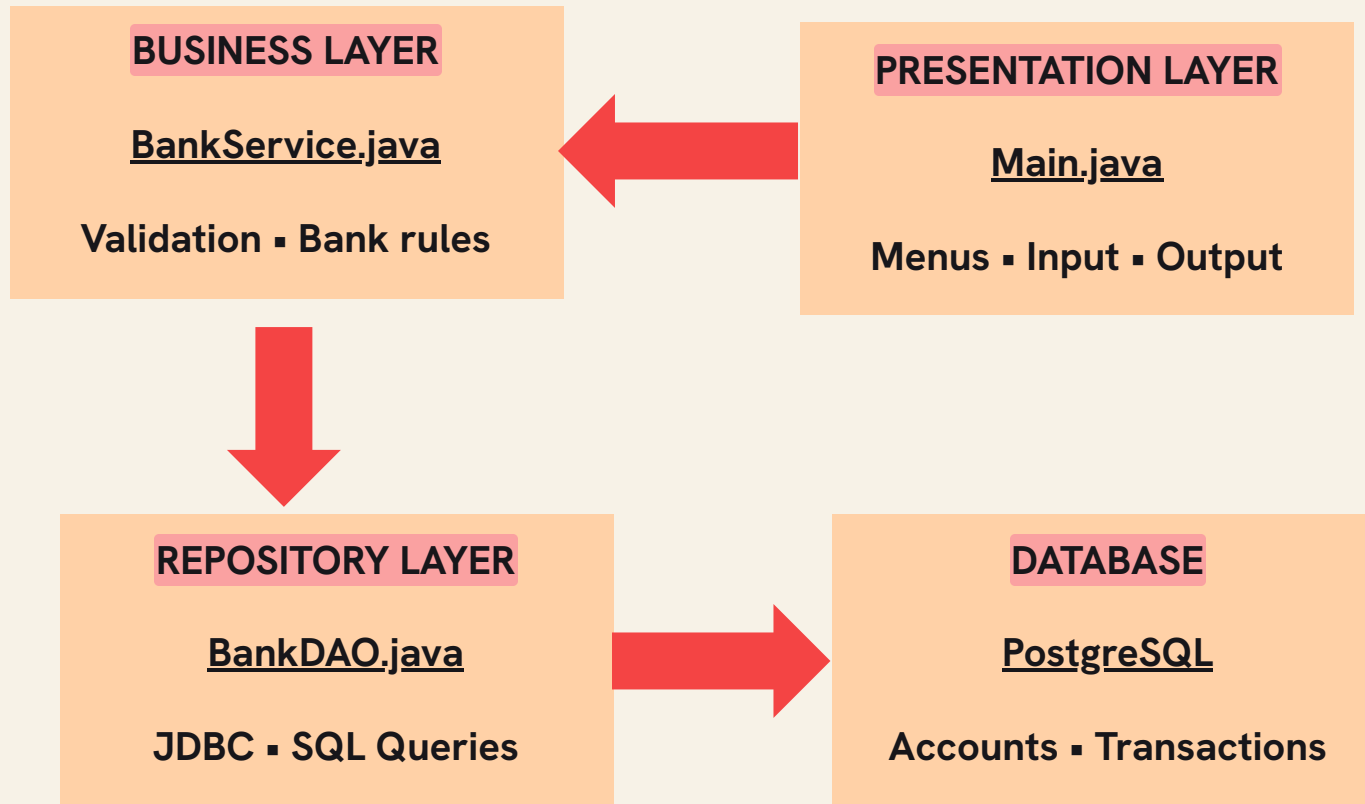
PostgreSQL



Git · GitHub

The Layered Architecture

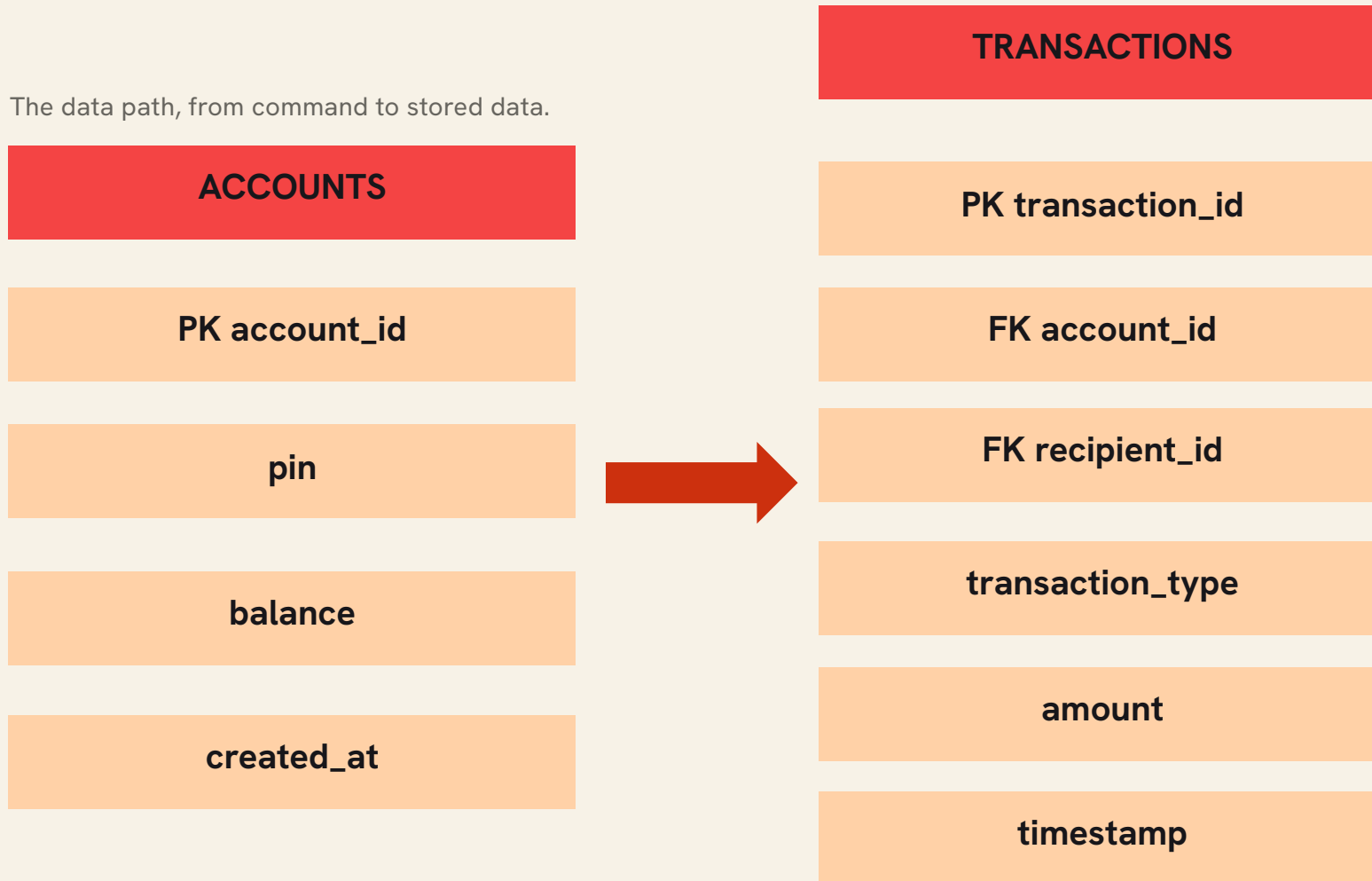
One application, distinct responsibilities.



```
bank-of-cli/  
├── src/  
│   ├── main/  
│   │   └── java/  
│   │       └── com/bankofcli/bankofcli/  
│   │           ├── Main.java  
│   │           ├── dao/  
│   │           │   └── BankDAO.java  
│   │           ├── model/  
│   │           │   ├── Account.java  
│   │           │   └── Transaction.java  
│   │           ├── service/  
│   │           │   └── BankService.java  
│   │           └── util/  
│   │               └── DatabaseConnection.java  
│   └── test/  
│       └── java/  
│           ├── AccDaoTest.java  
│           └── BankServTest.java  
├── pom.xml  
├── .gitignore  
└── README.md
```

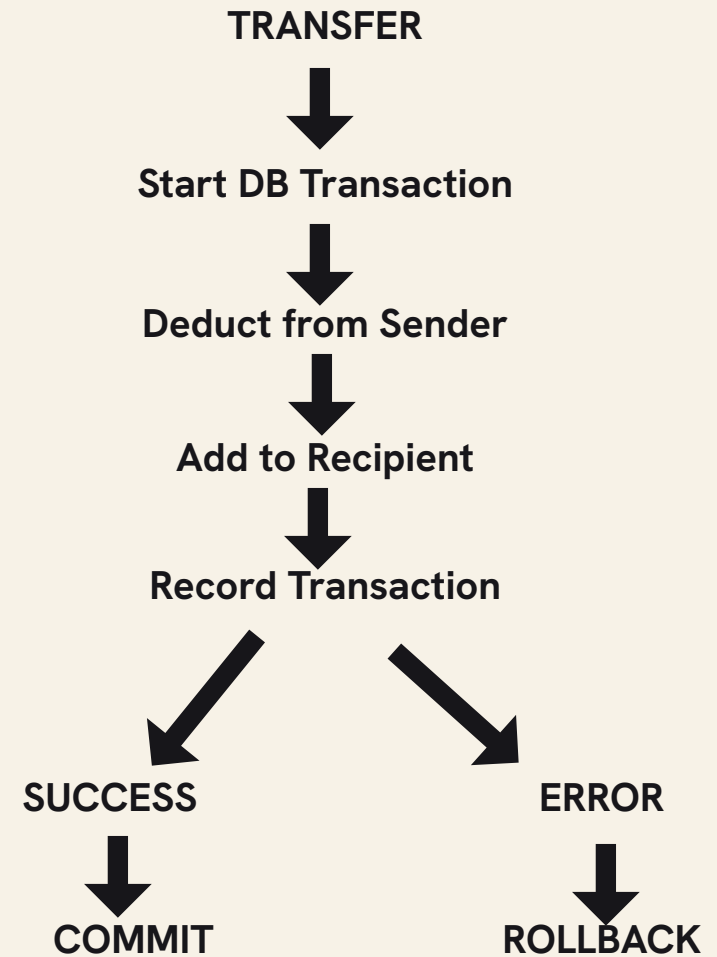
Database Design

The data path, from command to stored data.

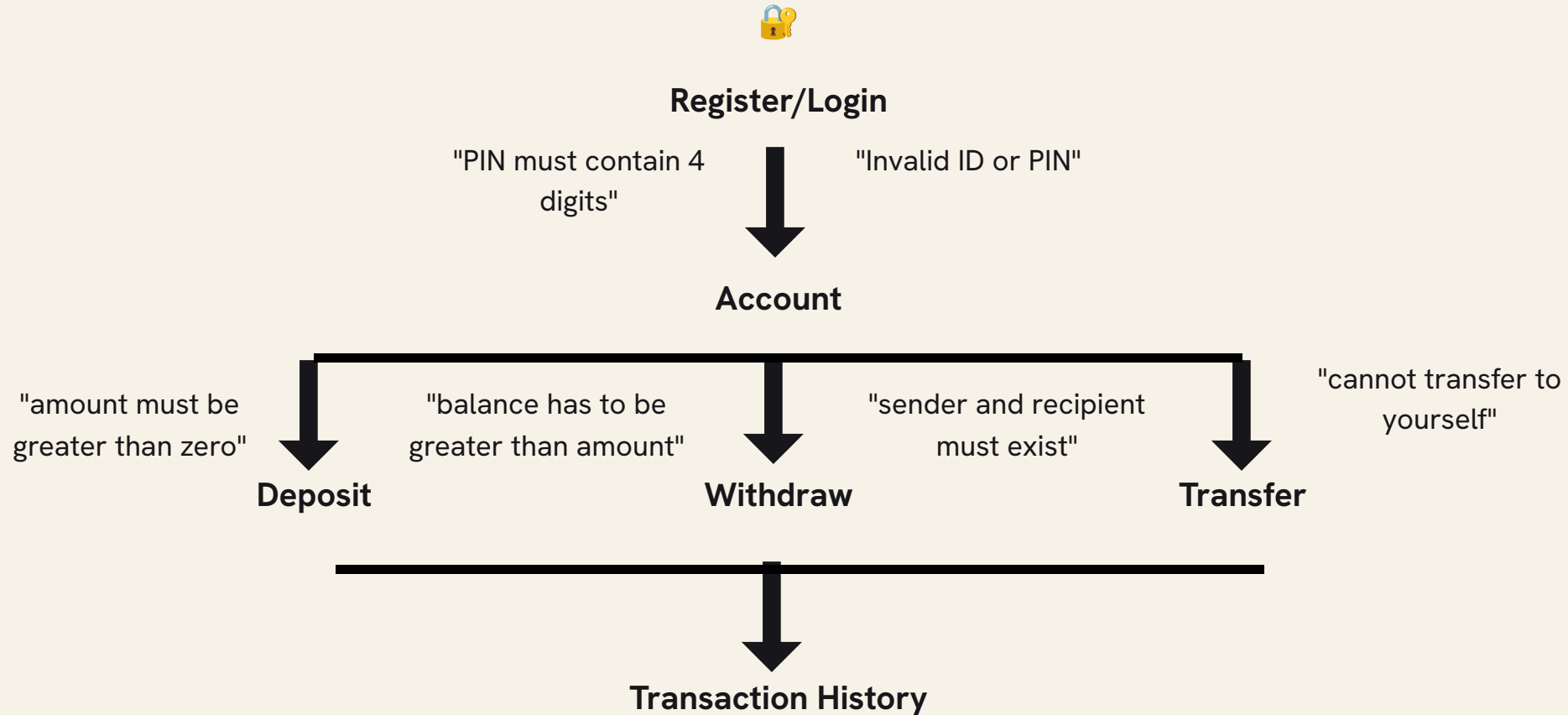


DATA
ACCESS
OBJECT

ATOMICITY
CONSISTENCY
ISOLATION
DURABILITY



Transaction Safety & Validation



Testing

SERVICE TESTS

JUNIT 5 + Mockito

Mock bankDAO

Test business rules

DAO TESTS

JUNIT 5 + PostgreSQL

Real Database

Test JDBC / SQL

Logging & Error Handling

```
2026-09-21 21:29:04 INFO com.bankofcli.bankofcli.dao.BankDAO - Account 1 authenticated successfully
2026-09-21 21:29:12 INFO com.bankofcli.bankofcli.dao.BankDAO - Deposit of 50 completed for account 1
2026-09-21 21:29:34 INFO com.bankofcli.bankofcli.dao.BankDAO - Withdrawal of 500 completed for account 1
2026-09-21 21:38:50 INFO com.bankofcli.bankofcli.dao.BankDAO - Account 1 authenticated successfully
2026-09-21 21:39:20 ERROR com.bankofcli.bankofcli.dao.BankDAO - Authentication failed for account 1
2026-09-22 09:34:49 INFO com.bankofcli.bankofcli.dao.BankDAO - Account 1 authenticated successfully
2026-09-24 08:46:06 INFO com.bankofcli.bankofcli.dao.BankDAO - Account 4 created successfully
2026-09-24 08:46:06 INFO com.bankofcli.bankofcli.dao.BankDAO - Deposit of 50.00 completed for account 4
2026-09-24 08:46:37 ERROR com.bankofcli.bankofcli.dao.BankDAO - Authentication failed for account 5
2026-09-24 08:46:44 INFO com.bankofcli.bankofcli.dao.BankDAO - Account 5 created successfully
2026-09-24 08:47:05 INFO com.bankofcli.bankofcli.dao.BankDAO - Account 5 authenticated successfully
2026-09-24 08:47:32 ERROR com.bankofcli.bankofcli.dao.BankDAO - Authentication failed for account 5
2026-09-24 08:47:39 ERROR com.bankofcli.bankofcli.dao.BankDAO - Authentication failed for account 5
2026-09-24 08:47:45 INFO com.bankofcli.bankofcli.dao.BankDAO - Account 5 authenticated successfully
2026-09-24 08:48:29 INFO com.bankofcli.bankofcli.dao.BankDAO - Deposit of 100.50 completed for account 5
2026-09-24 08:49:08 INFO com.bankofcli.bankofcli.dao.BankDAO - Withdrawal of 50.25 completed for account 5
2026-09-24 08:50:10 INFO com.bankofcli.bankofcli.dao.BankDAO - Transfer of 50 from account 5 to account 1 completed successfully
2026-09-24 08:50:37 INFO com.bankofcli.bankofcli.dao.BankDAO - Account 1 authenticated successfully
2026-09-24 10:41:55 ERROR com.bankofcli.bankofcli.dao.BankDAO - Authentication failed for account 1
2026-09-24 10:42:00 INFO com.bankofcli.bankofcli.dao.BankDAO - Account 1 created successfully
2026-09-24 10:42:05 INFO com.bankofcli.bankofcli.dao.BankDAO - Account 1 authenticated successfully
2026-09-24 10:44:25 ERROR com.bankofcli.bankofcli.dao.BankDAO - Authentication failed for account 4
```

User



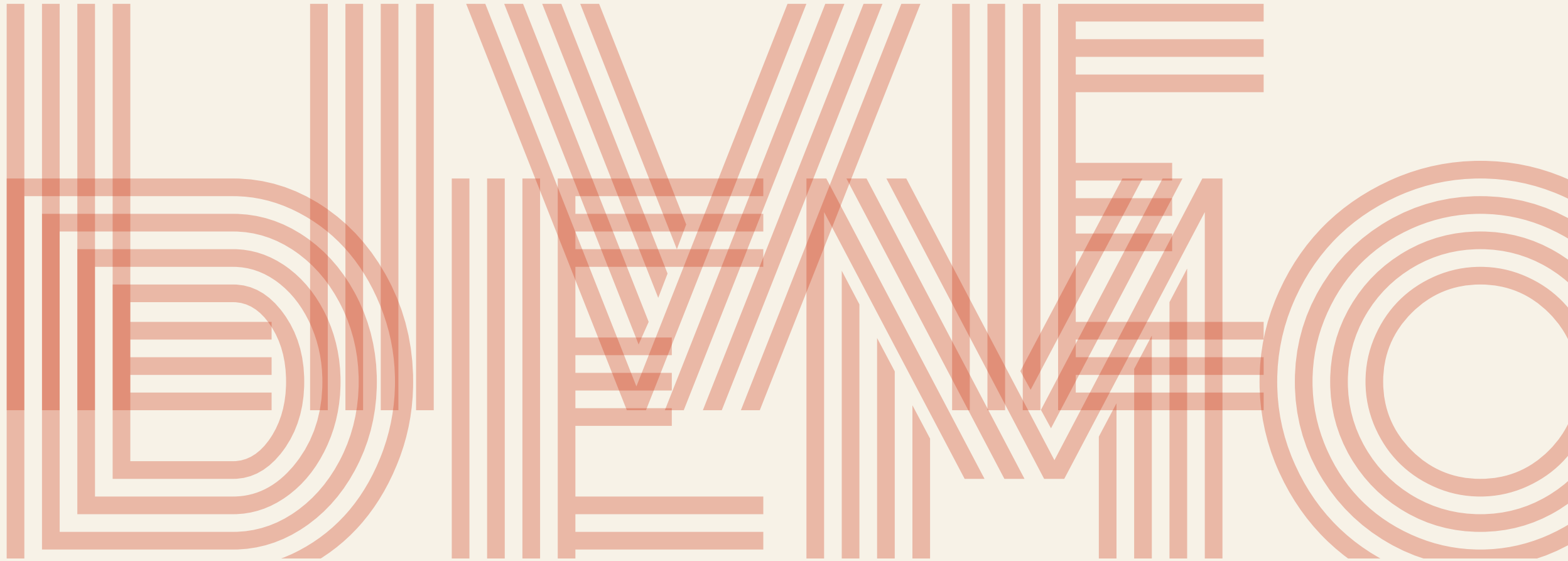
"Invalid account ID or PIN"



Developer / Log



Information about what happened



BANK OF CLI

Challenges and what I've learned

```

  BANK OF CLI

Account: #1
Balance: $1,000,000.00

[1] 🟢 View Balance
[2] + Deposit
[3] - Withdraw
[4] ⇄ Transfer
[5] 📄 Transaction History
[6] ↶ Logout
```

CHALLENGES

WHAT I LEARNED

Database transactions	→	Commit & rollback
Layer separation	→	Layered architecture
Testing dependencies	→	Mockito
Database persistence	→	JDBC + PostgreSQL
Error handling	→	Logging + exceptions